FROB C5 archived quotient Frobenius matrices

Validation audit, 22 October 2026; provenance clarification, 25 October 2026

Scope. The three frozen `RUN-FROB-91ee9c-7119f2` core cells use a monic degree-five quotient modulus g_q over $\mathbf{F}_q$. The target proves that the q-power image of each basis monomial $1, z, \ldots, z^4$ has exactly the five coordinates recorded in the source-bound replay matrix.

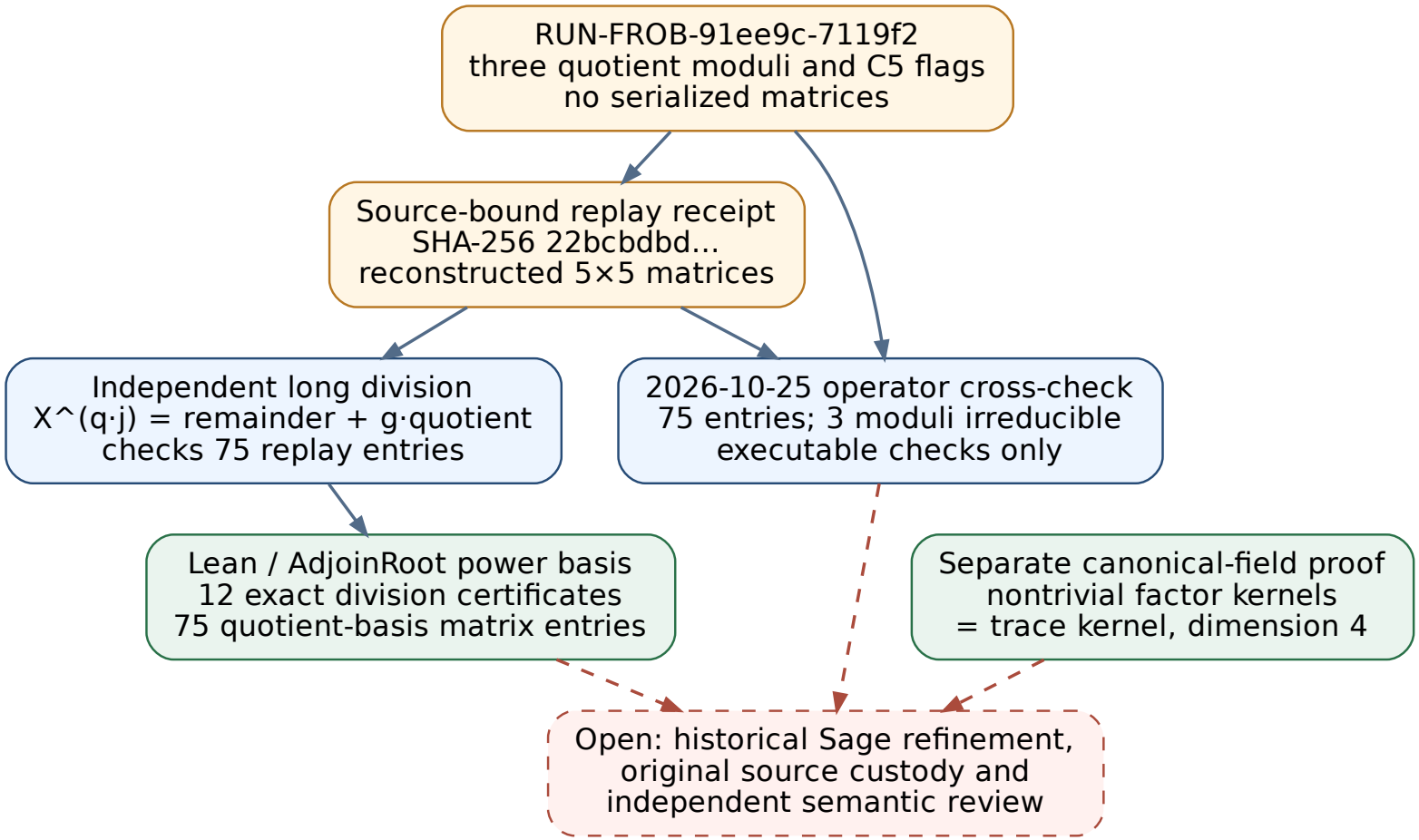
Base field	Frozen quotient modulus	Entries proved
$\mathbf{F}_{11}$	$z^5 + 2z^4 + 1$	5×5
$\mathbf{F}_{13}$	$z^5 + 8z^4 + 1$	5×5
$\mathbf{F}_{19}$	$z^5 + 3z^4 + 1$	5×5

Certificate construction. A standard-library Python generator reads the committed matrix replay receipt only after checking its SHA-256 and the replay checker's source hash. It independently divides each X^{qj} by g_q for $j = 1, \ldots, 4$, and requires the remainder coefficients to match the recorded matrix column. Lean checks all 12 explicit identities $X^{qj} = R_{qj} + g_q Q_{qj}$, the remainder degree bounds, and the constant column. Mathlib's `AdjoinRoot` power-basis representation then turns these into 75 exact matrix-entry equalities. The generator's `-check` compares generated Lean bytes; CRLF and receipt-hash mutations are rejected. An altered q11 matrix entry is also rejected after the test receipt is rehashed.

Machine verification. Source commit `c685c2951ac3` was checked with pinned Lean 4.35.0-rc1. The full 8,972-job build and 3,092-job target build passed; 625 theorem declarations used standard axioms only. The requested theorem's axiom audit passed. Independent semantic review is pending.

Provenance clarification, 25 October. The frozen run stores the quotient moduli and C5 flags, not serialized matrix entries. The replay reconstructs its matrices from those moduli. A separate read-only cross-check verified all 75 entries, seven source hashes, three raw-output hashes and executable irreducibility and ordering of the moduli. These checks do not certify the historical Sage execution or prove irreducibility in Lean.

Boundary. This proves the algebra of the archived quotient moduli and recorded matrices. It does not prove that the historical Sage driver implements the same operations on every path, establish original source custody, or supply independent semantic review. The separately checked canonical-field factor-kernel theorem is not identified with every historical execution by this certificate alone. C8 and the full experiment gate remain open. No scientific trial was rerun and no performance value changed.


RUN-FROB-91ee9c-7119f2
three quotient moduli and C5 flags
no serialized matrices
Source-bound replay receipt
SHA-256 22bcbdbd...
reconstructed 5×5 matrices
Independent long division
X^(q·j) = remainder + g·quotient
checks 75 replay entries
2026-10-25 operator cross-check
75 entries; 3 moduli irreducible
executable checks only
Lean / AdjoinRoot power basis
12 exact division certificates
75 quotient-basis matrix entries
Separate canonical-field proof
nontrivial factor kernels
= trace kernel, dimension 4
Open: historical Sage refinement,
original source custody and
independent semantic review